# Integrity in Digital Scholarly Editing: The GreekSchools Case

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Textual scholarship aims to reconstruct and publish texts by investigating philological phenomena and documenting findings through critical apparatuses or editorial notes (Stussi 2015). Textual scholarship involves intricate, multi-stage processes that are further complicated by the digital environment, which demands formal protocols and standardized workflows (Shillingsburg 2015; Berti 2019). This formalization can be burdensome and error-prone for traditional scholars (Boschetti and Del Grosso 2020) depending on the amount of text and textual phenomena that are in place.The gradual transition to Digital Scholarly Editions (DSE) offers benefits, including explicit encoding of textual phenomena in a machine-actionable format for example, by complying to the XML/TEI encoding guidelines (Del Grosso et al. 2023; Driscoll and Pierazzo 2016), and the ability to conduct thorough automated text searches. In this context, a strong dichotomy between traditional and digital approaches to scholarly editing may appear to exist.

The DSL-based DSE method (Zenzaro et al. 2023; Zenzaro et al. 2021) establishes a middle ground approach. At its core, this method aims to preserve scholars' traditional ecdotic knowledge while utilizing computational capabilities through the use of Domain-Specific Languages (DSLs). This bridges the gap between traditional methodologies and the digital techniques. Furthermore, the DSL-based DSE method facilitates the implementation of open science principles, particularly the FAIR principles (Wilkinson et al. 2016).

The DSL-Based DSE method derives from the Euporia annotation method (Boschetti et al. 2023) and adheres to the following principles: 1) enabling scholars to work in a familiar text-focused environment, 2) ensuring machine actionability and interoperability (Fig 1), 3) granting the overall consistency of the edition, and 4) improving the coherence of each statement.

The first principle is accomplished by employing a set of DSLs that closely align with traditional editorial conventions, making the text easily manageable for scholars while also ensuring it is represented in a machine-actionable format.

The second principle is realized by integrating computational functionalities into the text, interpreted as a DSL, that adhere to shared standards, such as XML/TEI converters and the Web Annotation Data Model format for annotations.

The third principle involves consistency: all parts of a digital scholarly edition must be non-contradictory with respect to one another. For instance, a gap encoded as a 5-letters lacuna in the diplomatic edition cannot be filled by a 7-letters conjecture in the related critical edition.

The fourth principle concerns coherence: all the same phenomena must be encoded in the same way and all different phenomena must be encoded in different ways. This principle prevents (or at least inhibits) the introduction of false positives and false negatives and consequently optimizes the precision and recall of the overall system.

The GreekSchools ERC-885222 project provides an ideal environment for validating the DSL-based DSE method. The project's primary objective is to publish scholarly editions of selected Herculaneum papyri, which would comprise over 1,800 carbonized scrolls containing unedited Greek philosophical texts (Ranocchia, 2022). The potential of computational advancements to improve papyrologists' workflows has been demonstrated by recent AI challenges and initiatives (Riemenschneider and Anette 2023), including tools like Ithaca (Assael et al. 2022) and the Vesuvio Challenge, as well as platforms such as the well-known papyri.info. Moreover, the GreekSchools project aligns with the goals of the Humanities and Heritage Italian Open Science Cloud (H2IOSC) consortium[1] targeting CLARIN[2], DARIAH[3]; E-RIHS[4], and OPERAS[5] Research Infrastructures (RIs).

In the context of the GreekSchools project we are developing CoPhiEditor: a web-based digital

scholarly environment for Digital Scholarly Editing. This tool fits into the set of initiatives such as Proteus (Williams et al. 2015), Textual Communities (Robinson and Bordalejo 2016), and LEAF-writer[6].

The primary objective of CoPhiEditor is to enable a collaborative and cooperative workflow, allowing multiple scholars to engage with the same text simultaneously or to work on different sections asynchronously. By building upon this workflow and its underlying principles, CoPhiEditor utilizes formal Domain-Specific Languages (DSLs), language technologies, and the IIIF framework to provide automated support, thereby enhancing the overall quality of the editing process and the final editorial output. CoPhiEditor significantly minimizes the need for manual checks to ensure compliance with editorial conventions (Fig. 4), such as encoding lacunae, unclear characters, and readings from apographs. It also verifies cross-textual constraints, ensuring consistency between diplomatic and literary transcriptions as well as accurate references in apparatus entries. Furthermore, it optimizes the time dedicated to critical textual phenomena, such as assessing hypotheses for reconstructing texts with variant readings and conjectures. Additionally, CoPhiEditor adheres to standard formats like EpiDoc (Fig. 5), which ensures interoperability of the digital edition across various software applications. This capability facilitates diverse processing tasks, including search functionalities, information extraction, and the creation of training sets for machine learning applications.

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Figure 1 - DSL-Based DSE approach: editing with editorial convention DSL (on top) and TEI/EpiDoc automatic generated document (at bottom)

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Figure 2 - CoPhiEditor main GUI

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Figure 3 - Comments in CoPhiEditor

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Figure 4 - Automatic validation for editorial conventions

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Figure 5 - TEI/EpiDoc of an editorial deletion and addition

# Notes

1. https://www.h2iosc.cnr.it
2. https://www.clarin.eu/
3. https://www.dariah.eu/
4. https://www.e-rihs.eu/
5. https://operas-eu.org/
6. https://leaf-writer.leaf-vre.org/#about

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